Machine Learning Approaches to Predicting Cancer Cell Line Sensitivity to Ulixertinib

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***Abstract*—Predicting drug response for cancer cells is a key challenge in the field of precision medicine. In this study, machine learning models were used in attempt to classify the sensitivity of cancer cell lines to ulixertinib using data from the Genomics of Drug Sensitivity in Cancer (GDSC) dataset. The problem was formulated as a binary classification task, with ln(IC_{50}) values being used to label lines as sensitive or resistant. Logistic regression and random forest models were trained and subsequently evaluated using an 80:20 train-test split and 10-fold cross-validation. Both models achieved moderate and similar prediction performance, with cross-validation accuracies of roughly 66-67%. Feature analysis revealed that a mix of biological and experimental variables contributed to predictions. Whilst the results demonstrate the potential of machine learning for predicting drug response—especially in the field of oncology—several limitations signify the need for further research prior to clinical application.**

Keywords—Precision medicine, drug response prediction, IC_{50}, cancer cell lines, ulixertinib, machine learning, binary classification.

I. Introduction

Precision medicine—defined as tailoring healthcare to a person's genetics, lifestyle, and environment [1]—could prove to be an invaluable tool to bettering patient prognosis for a plethora of conditions.

Given that cancer is one of the leading causes of mortality and was responsible for roughly 10 million deaths globally in 2020 [2], it is clear that oncology would be an apt area to target in this fashion.

One promising application of precision medicine is the ability to predict whether a patient with cancer will respond to a drug prior to it being deployed. If a patient is forecasted to respond poorly to a drug based on available biomarkers, it prevents unnecessary side effects and crucial time and money from being wasted on treatment that is unlikely to be effective. Congruently, this could be extended to comparing multiple drugs to select which is likely to be optimal.

One method for implementing this concept is using supervised machine learning methods, which were the focus of this study. The problem was formulated as a binary classification task, with cancer cell lines being classified as either sensitive or resistant. The data for this task was obtained from the Genomics of Drug Sensitivity in Cancer (GDSC) dataset, which contained drug response data and genomic profiles of cancer cell lines.

This study provides a simplified and interpretable approach to drug response prediction via classification with a single drug. In this study, cancer cell line sensitivity to ulixertinib was predicted using Logistic Regression and Random Forest models. This was followed by evaluation and feature analysis, the latter being used to compare feature contributions across models.

II. Related Works

Previous studies have demonstrated both that drug response is dependent on cancer cell biology and that machine learning models can function as effective tools for predicting drug response in cancer cell lines.

Importantly, Garnett et al. [3] showed that cancer cell genes can function as drug sensitivity biomarkers, supporting the use of such features for predictive modelling. Garnett et al.'s paper generated pharmacogenomic data similar in nature to that found in the GDSC dataset, with hundreds of cancer cell lines screened and 130 drugs tested on them.

Furthermore, Menden et al. [4] utilised machine learning models to predict cancer cell line response to drug treatment, which is highly relevant to the work outlined in this paper. This work was distinct in two important ways, however. Firstly, they performed the work as a regression task, predicting IC_{50} values directly, which may be less interpretable for a single drug in a clinical context. Moreover, their model was built on both genomic features of cell lines and the given drug's chemical properties. This two-pronged approach increases model complexity and the interplay between the two factors makes it potentially difficult to assess the contribution of each feature, thereby making it harder to draw conclusions from the results.

Xia et al. [5] provided promising evidence that models, including random forest, trained on a single cancer cell line-based dataset (including GDSC) to predict drug response generalise fairly well to other available, similar datasets. This was necessary as cross-validation, although an important initial test of model accuracy, is likely overoptimistic for estimating predictive performance for other independent test sets, with success in the latter being necessary for real-world predictions involving multiple datasets. Importantly, this study was limited to cell-line data and we cannot conclude that the results would be similar with real patient data.

III. Methodology

The following section describes how binary classification of ulixertinib sensitivity was carried out, including data selection, preprocessing, model development, and evaluation.

A. GDSC Dataset and Drug Selection

The GDSC dataset was downloaded from Kaggle. It was comprised of several drug response metrics for many drugs across different cancer cell lines alongside biological and experimental features.

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The drug response metrics provided were ln(IC_{50}), AUC, and Z-score. This dataset was ideal for machine learning modelling of this nature as both input features and a target variable to predict (drug sensitivity) were present.

Due to time constraints and to maintain a relatively simple and focused scope for the work, modelling was based on a single drug. Ulixertinib was selected for this task as it had the largest number of observations in the dataset whilst providing data with suitable characteristics for modelling, such as relatively low levels of missing data. A new dataset containing only ulixertinib data was subsequently constructed. The final dataset for ulixertinib consisted of 1609 cancer cell line samples.

B. *Exploratory Data Analysis*

Upon narrowing to ulixertinib, the distribution of the target variable (ln(IC_{50})) was visualised using a histogram to assess its distribution, which appeared to be approximately symmetric as shown in figure 1.

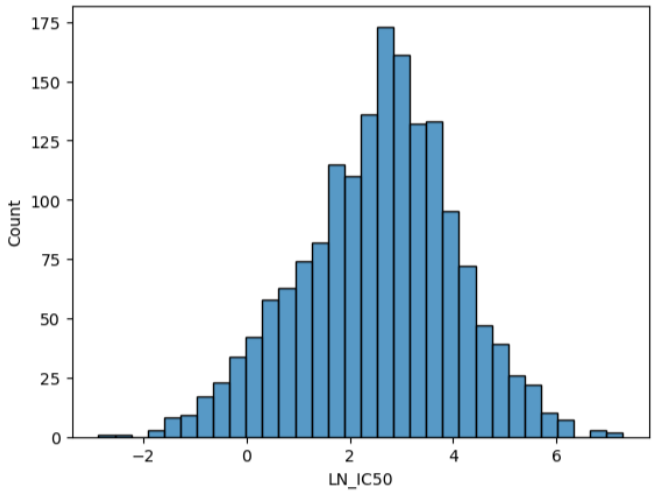


Figure 1. A histogram showing the roughly symmetric distribution of the target variable, ln(IC_{50}).

The data types and number of missing values in each column were also analysed.

C. *Target Variable Creation*

Drug sensitivity was quantified as ln(IC_{50}), a log-transformed measure of the concentration of a drug required for 50% inhibition which is a widely used indicator of drug potency in literature [3].

To facilitate binary classification, ln(IC_{50}) values were binarised by comparing them to a median threshold value. Values below the threshold were assigned 1 (sensitive) and those above it were assigned 0 (resistant). This ensured class distribution was balanced and the use of a median threshold was supported by the roughly symmetric distribution of the target variable.

D. *Data Preprocessing*

Columns that were redundant for model building, such as identifiers and drug metadata, were removed from the dataset.

Upon discovering high correlation between the target variable and AUC/Z-score, the latter two features were also removed to prevent data leakage.

A relatively small number of missing values were present in multiple columns of the dataset, however one contained around 21.3% missing values (Cancer Type (matching TCGA label)). Henceforth, missing data was handled by removing this entire column and removing all remaining rows with missing data. The presence of related features such as TCGA description helped to justify dropping of the entire column.

The categorical features in the dataset were one-hot encoded to convert them into numerical data that models can be trained on.

E. *Train/test Split*

The dataset was split into training and testing sets using an 80:20 split, respectively. The testing data is held out from model training so it can be used as unseen data to evaluate models and assess their generalisability.

F. *Training Models for Binary Classification*

Two supervised machine learning models were trained for the binary classification task: logistic regression and random forest. Logistic regression was chosen to act as a baseline, linear model. Both models were trained on the training dataset and with default parameters.

G. *Evaluation*

The performance of both models was initially evaluated on the test set. Accuracy was used as the primary metric (due to balanced data) and confusion matrices were also generated to provide additional insight into performance.

The performance of both models was then assessed using 10-fold cross-validation on the training data to obtain a more robust measure of predictive performance and ensure previous results were not dependent on a particular data split.

A paired t-test was conducted on the cross-validation accuracies of both models to test whether there was a statistically significant difference between the mean accuracies.

Analysis of standardised coefficients was also conducted for logistic regression as well as feature importance for random forest to assess which features contributed the most to predictions.

IV. Results

The results for the key aforementioned experiments are listed below, highlighting that moderate and reliable classification accuracy was achieved with both of the implemented models.

A. *Model Performance Evaluation*

The performance of both models was initially evaluated on the test set, with an accuracy of 71.43% for logistic regression and 72.98% for random forest.

The performance of logistic regression and random forest was also tested via 10-fold cross-validation, with moderate mean accuracies of 66.19% and 67.24% achieved,

respectively (2dp.). These were slightly lower but fairly consistent with the initial accuracy derived from the test set, suggesting the test performance was not substantially overestimated.

The result of the paired t-test indicated that the higher cross-validation accuracy for random forest was not statistically significant ($p = 0.47$ (2dp.), $p > 0.05$).

Confusion matrices were used to further assess the performance of classification. Both models demonstrated a fairly balanced distribution of true and false predictions as shown for logistic regression in figure 2.

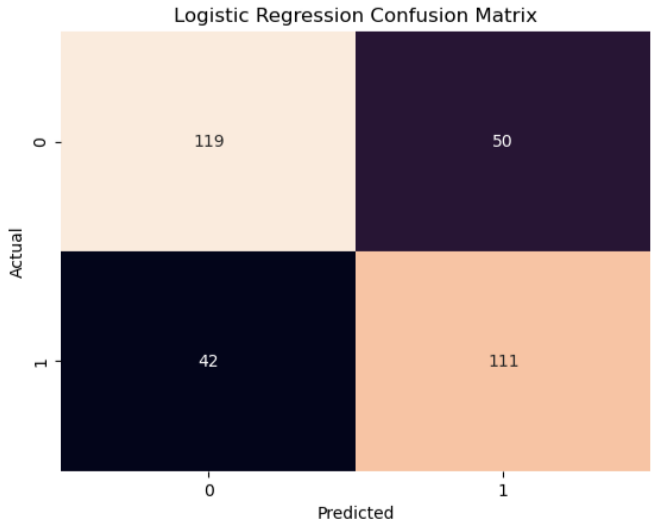


Figure 2. A confusion matrix for the logistic regression model showing a roughly balanced split of true positive/true negative and false positive/false negative predictions.

B. Coefficient/Feature Importance Analysis

Standardised coefficients and feature importances were analysed for logistic regression and random forest, respectively. Some of the most impactful features for each model overlapped but many were notably distinct. The results for both models showed a mix of experimental and biological features, as shown in figures 3 and 4.

TCGA_DESC_KIRC	0.562169
GDSC Tissue descriptor 2_myeloma	0.516188
TCGA_DESC_BRCA	0.514640
TCGA_DESC_HNSC	0.494194
GDSC Tissue descriptor 2_head and neck	0.489790
Growth Properties_Semi-Adherent	0.406469
TCGA_DESC_CESC	0.358554
TCGA_DESC_PRAD	0.334834
TCGA_DESC_LUAD	0.326197
GDSC Tissue descriptor 2_leukemia	0.318908

Figure 3. The top 10 highest standardised coefficients for the logistic regression model.

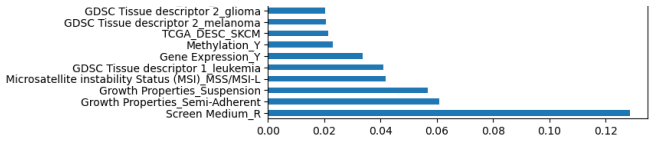


Figure 4. The top 10 most important features for the random forest model.

V. Discussion

Before discussing what the potential applications of the results of this study are, it is important to outline limitations faced despite a moderate accuracy of prediction being achieved.

Firstly and likely most importantly, binary classification of a cell line as either sensitive or resistant to a treatment is not an accurate reflection of how a drug would function in a clinical setting. There is no single IC_{50} value where a drug suddenly becomes effective for a patient and the decision on whether a drug should be given should be based on numerous factors i.e. researched biological thresholds.

An additional distinction that must be made between this research and clinical application is the fact that this study used data from cancer cell lines (in vitro) and cancer cells may behave differently in real-world disease cases inside patients (in vivo).

One major area that future works could target is attempting to reproduce similar results using data acquired from patient cancer cells as this would provide a much more reliable indicator of whether this methodology could be useful in a clinical setting. Acquiring data of this nature is likely to be costly and invasive, however. Another would be improving the results of this study via techniques like comparing imputation methods for missing data, tuning hyperparameters, or implementing other available machine learning methods e.g. deep learning. Baptista et al. [6] have already shown that deep learning could be a promising alternative to traditional machine learning in this field. The methods could also be extrapolated to other drugs to determine whether similar accuracies can be obtained to that demonstrated in this paper for ulixertinib. Reformulation of the problem to a regression-based task for predicting IC_{50} values could be used in this case to compare multiple drug's efficacies on the same cell line based on its biological properties.

Another observation worth commenting on is the implication of the results of analysing feature impact to the prediction of logistic regression and random forest. A mixture of experimental and biological features were important for both models, potentially reducing the biological interpretability of the results as experimental conditions which would be absent in real world treatment are likely playing a part in model learning and predictions to an extent. However, being able to show the biological features that have the biggest impact on predictions could provide useful insight on which biomarkers clinicians may benefit from obtaining data on. This is a benefit to using more interpretable models such as logistic regression and random forest over a deep neural network, for example.

Furthermore, The high similarity between classification accuracies for both models suggests that in this case, increasing model complexity does not significantly improve predictive performance.

Congruent with Xia et al.'s findings [5], the consistency between the test set and cross-validation results suggests that the models generalise fairly well to unseen data, although this may be limited to similar cell line datasets.

Prior to clinical application, prioritising either recall or precision depending on the treatment objective could also be an important area to target instead of simply just using

accuracy. Prioritising recall (mitigating false negatives) would likely mean fewer effective treatments would be missed, whereas prioritising precision (mitigating false positives) would likely result in fewer patients receiving ineffective treatment and unnecessary side effects.

Despite the aforementioned limitations and observations, the results of this study show that biological features can be used to classify cell line vulnerability to a specific drug with moderate accuracy. This suggests that while meaningful patterns do exist within the GDSC data, drug response prediction remains a challenging problem and is far from being reliable enough for clinical decision-making. As with most machine learning problems, having more available data would likely be invaluable and investing time and money into researching biomarkers further could prove to be very useful.

VI. Conclusion

The results of this study act as initial proof of concept that cancer cell line genes can act as biomarker data, which machine learning models can then be trained on to predict drug efficacy with moderate accuracy. The study was not without several notable limitations and further research and adaptation in this area is likely to be very useful prior to clinical implementation, with machine learning models hopefully one day functioning alongside clinicians to facilitate more informed decision-making in the field of oncology as well as medicine as a whole.

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